dependent on quantum mechanics from the perspective of one of the other string theories (for example, a process whose detailed numerical properties are influenced by quantum considerations but whose qualitative form is similar to what it would be in a purely classical world). This means that quantum mechanics is thoroughly intertwined within the duality symmetries underlying string/M-theory: They are inherently quantum-mechanical symmetries, since one of the dual descriptions is strongly influenced by quantum considerations. This indicates forcefully that the complete formulation of string/M-theory—a formulation that fundamentally incorporates the newfound duality symmetries—cannot begin classically and then undergo quantization, in the traditional mold. A classical starting point will necessarily omit the duality symmetries, since they hold true only when quantum mechanics is taken into account. Rather, it appears that the complete formulation of string/M-theory must break the traditional mold and spring into existence as a full-fledged quantum-mechanical theory.

Currently, no one knows how to do this. But many string theorists foresee a reformulation of how quantum principles are incorporated into our theoretical description of the universe as the next major upheaval in our understanding. For example, as Cumrun Vafa has said, "I think that a reformulation of quantum mechanics which will resolve many of its puzzles is just around the corner. I think many share the view that the recently uncovered dualities point toward a new, more geometrical framework for quantum mechanics, in which space, time, and quantum properties will be inseparably joined together."5 And according to Edward Witten, "I believe the logical status of quantum mechanics is going to change in a manner that is similar to the way that the logical status of gravity changed when Einstein discovered the equivalence principle. This process is far from complete with quantum mechanics, but I think that people will one day look back on our epoch as the period when it began."6

With guarded optimism, we can envision that a reframing of the principles of quantum mechanics within string theory may yield a more powerful formalism that is capable of giving us an answer to the question of how the universe began and why there are such things as space and time—a formalism that will take us one step closer to